Email Deliverability Checker

Software Development Course Project Submission

Project Title:	Email Deliverability Checker (Non-Intrusive Mailbox Verification)
Author / Student:	Student Course Submission
Live Application URL:	*[Pending Deployment - e.g., https://email-deliverability-checker.onrender.com]*
GitHub Repository URL:	*[Pending Publication - e.g., https://github.com/<username>/email-deliverability-checker]*
Technology Stack:	Python 3.12, Flask, dnspython, Gunicorn, Vanilla JS, CSS3, HTML5

1. Project Summary

The Email Deliverability Checker is a complete web application designed to evaluate the delivery risk of batches of up to 100 email addresses without sending actual emails. It categorizes addresses into 'Likely deliverable', 'Likely to bounce', or 'Unknown'.

Critical Safety Guarantee: The application never issues the SMTP DATA command, never initiates bulk campaigns, and never stores user email lists in any database or disk logs.

2. Key Features

- Multi-Stage Inspection: RFC 5322 syntax validation, DNS MX queries, and safe SMTP recipient checks.
- Modern & Polished Interface: Clean white/indigo styling, live counter, summary cards, and responsive tables.
- Safe Bounded Concurrency: Deduplicates lists and verifies up to 100 emails concurrently with ThreadPoolExecutor.
- CSV Export: Instant one-click export of structured diagnostic deliverability reports.
- Transparent Reporting: Provides clear explanations for why every result was classified.

3. Deliverability Classifications

Category	Technical Criteria	Practical Interpretation
Likely deliverable	Valid syntax, active MX, and SMTP 250 OK accepted.	Probable inbox delivery (subject to anti-spam).
Likely to bounce	Malformed syntax, NXDOMAIN, or SMTP 550 rejection.	Address is defective or nonexistent. High bounce risk.
Unknown	Valid MX exists, but server hides status / port 25 blocked	Cannot confirm mailbox existence without sending.

4. Limitations & Cloud Hosting Realities

- Catch-All Mail Servers: Many corporate domains accept all addresses at the edge, concealing mailbox validity.
- Cloud Port 25 Blocking: Platforms like Render, Heroku, and AWS block outbound port 25 on free tiers.
  When deployed on Render, the system safely verifies DNS MX records and marks addresses as 'Unknown' with an explanation.

5. Automated Test Results

Automated test suite (pytest) passed: 22 passed in 1.62s (100% pass rate).
All tests use mocked DNS and SMTP calls, strictly verifying zero email transmission and zero DATA commands.